BST Working Paper — Volume 5: The Predictions — Chapter 1: Experimental Predictions and Research Program

2026-05-19 (Tuesday volume:chapter reorganization)

43. Experimental Predictions and Falsifiability

43.1 The Economy of the Framework

BST has three structural inputs: a 2D substrate with S^2 topology, an S^1 communication fiber, and the requirement that the resulting contact graph be self-consistent. From these inputs the framework derives the bounded symmetric domain D_{IV}^5 as the configuration space, the channel capacity 137, and Haldane exclusion statistics with parameter $g = 1/137$. Everything else follows. There are no free parameters to adjust, no compactification geometries to choose, no landscape of vacua to navigate. The predictions either match observation or the framework is wrong. Few moving parts means few places to hide.

43.2 Parameter-Free Predictions (Established)

Prediction	BST Value	Observed	Status
Fine structure constant α^{-1}	137.036082 (Wyler, D_{IV}^5 volume)	137.035999 (CODATA)	✓ 0.0001%
α^{-1} independent derivation	137.035 (cost function + Bergman mixing, D_{IV}^5 mode density)	137.036 (Wyler)	✓ 5 ppm
Cosmological constant Λ	$F_{\text{BST}} \times \alpha^{56} \times e^{-2} = 2.8994 \times 10^{-122}$	2.90×10^{-122} Pl	✓ 0.02%
Vacuum free energy F_{BST}	$\ln(N_{\text{max}} + 1)/(2n_C^2) = \ln(138)/50 = 0.098545$	0.09855 (partition fn)	✓ exact
Committed contact scale d_0/ℓ_{Pl}	$\alpha^{2(n_C+2)} \times e^{-1/2} = \alpha^{14} \times e^{-1/2} = 7.365 \times 10^{-31}$	7.37×10^{-31} (from observed Λ)	✓ 0.005%
Phase transition temperature T_c	$N_{\text{max}} \times 20/21 = 130.476$ BST units	130.5 BST units (partition fn)	✓ 0.018%
GUT coupling N_{GUT}	$4\pi^2 \approx 39.48$	~ 40 (1.3%)	✓
Weinberg angle $\sin^2\theta_W$	$N_c/(N_c + 2n_C) = 3/13 = 0.23077$	0.23122 (MS-bar)	✓ 0.2%
W boson mass m_W	$m_Z\sqrt{10/13} = 79.977$ GeV	80.377 GeV	✓ 0.5%
Strong coupling $\alpha_s(m_p)$	$(n_C + 2)/(4n_C) = 7/20 = 0.350$	~0.35	✓ ~0%
Strong coupling $\alpha_s(m_Z)$	Geometric β-function, $c_1 = 3/5$	0.1175	✓ 0.34%
Baryon asymmetry η_b	$(3/14)\alpha^4 = N_c/(2g) \times \alpha^4$ (T929)	$(6.104 \pm 0.058) \times 10^{-10}$	✓ 0.45%
Hubble constant H_0	67.29 km/s/Mpc (Route C: full CAMB, Toy 677)	67.36 ± 0.54 (Planck)	✓ 0.1%
Neutrino mass m_{ν_3}	$(10/3)\alpha^2 m_e^2/m_p = 0.0494$ eV	≈ 0.050 eV	✓ 1.8%
Neutrino mass m_{ν_2}	$(7/12)\alpha^2 m_e^2/m_p = 0.00865$ eV	≈ 0.00868 eV	✓ 0.35%
Neutrino mass m_{ν_1}	0 (exactly, Z_3 Goldstone)	< 0.009 eV	✓ prediction

Prediction	BST Value	Observed	Status
Solar splitting Δm^2_{21}	$m^2_{\nu_2} = 7.48 \times 10^{-5}$ eV^2	$(7.53 \pm 0.18) \times 10^{-5}$ eV^2	✓ 0.7%
Mass ratio m_{ν_3}/m_{ν_2}	$40/7 = 5.714$	≈ 5.79	✓ 1.4%
PMNS $\sin^2\theta_{12}$	$(3/10)(44/45) = 0.2933$ (T1446)	0.2935 ± 0.012	✓ 0.06%
PMNS $\sin^2\theta_{23}$	$(4/7)(44/45) = 0.5587$ (T1446)	0.561 ± 0.018	✓ 0.40%
PMNS $\sin^2\theta_{13}$	$1/(N_c^2 n_C) = 1/45 = 0.02222$	0.02203 ± 0.00056	✓ 0.9%
CKM Cabibbo angle $\sin\theta_C$	$2/\sqrt{79} = 0.22502$ (T1444: $\text{rank}^4 n_C - 1 = 79$)	0.22501 ± 0.00068	✓ 0.004%
CKM CP phase γ	$\arctan(\sqrt{n_C}) = \arctan(\sqrt{5}) = 65.91°$	$65.5° \pm 2.5°$	✓ 0.6%
Wolfenstein $\bar{\rho}$	$1/(2\sqrt{2n_C}) = 1/(2\sqrt{10}) = 0.158$	0.159 ± 0.010	✓ 0.6%
Wolfenstein $\bar{\eta}$	$1/(2\sqrt{2}) = 0.354$	0.349 ± 0.010	✓ 1.3%
Jarlskog invariant J_{CKM}	$\sqrt{2}/50000 = 2.83 \times 10^{-5}$; $50000 = n_C^5 \times (2^{\text{rank}})^2$	$(2.77 \pm 0.11) \times 10^{-5}$	✓ 2.1%
CKM $\lVert V_{ub} \rVert$	$A\lambda^3/\sqrt{C_2} = 1/(50\sqrt{30}) = 0.003651$	0.003660 ± 0.000110	✓ 0.25%
Higgs quartic λ_H	$\sqrt{2/n_C!} = 1/\sqrt{60} = 0.12910$	0.12938 (from m_H)	✓ 0.22%
Higgs mass (Route A)	$v\sqrt{2\sqrt{2/5!}} = 125.11$ GeV	125.25 ± 0.17 GeV	✓ 0.11%
Higgs mass (Route B)	$(\pi/2)(1-\alpha)m_W = 125.33$ GeV	125.25 ± 0.17 GeV	✓ 0.07%
Fermi scale v (Higgs vev)	$m_p^2/(g \cdot m_e) = 36\pi^{10} m_e/7 = 246.12$ GeV, g=7=genus	246.22 GeV	✓ 0.046%
W boson mass m_W (Route B)	$n_C m_p/(8\alpha) = 80.361$ GeV	80.377 GeV	✓ 0.02%
Top quark mass m_t	$(1-\alpha)v/\sqrt{2} = 172.75$ GeV	172.69 ± 0.30 GeV	✓ 0.037%
Number of colors N_c	3 (from Z_3 center)	3	✓
Baryon = 3 quarks	Required by Z_3 closure	Observed	✓
Proton charge radius r_p	$\dim_{\mathbb{R}}(\mathbb{CP}^2)/m_p = 4\hbar/(m_p c) = 0.8412$ fm	0.84075 ± 0.00064 fm (CODATA)	✓ 0.058%
Proton/electron mass ratio	$(n_C+1)\pi^{n_C} = 6\pi^5 = 1836.118$	1836.153	✓ 0.002%
Muon/electron mass ratio	$(24/\pi^2)^6 = [K_3(0,0)/K_1(0,0)]^{\dim_{\mathbb{R}}(D^3_{IV})} = 206.761$	206.768	✓ 0.003%
Tau/electron mass ratio	$(24/\pi^2)^6 \times (7/3)^{10/3} = 3483.8$	3477.2	✓ 0.19%
Quark ratio m_s/m_d	$4n_C = 20$	$20.0 \pm \sim 5\%$	✓ $\sim 0\%$
Quark ratio m_t/m_c	$N_{\max} - 1 = 136$	$135.98 \pm \sim 1\%$	✓ 0.017%
Quark ratio m_b/m_τ	genus$/N_c = 7/3 = 2.333$	$2.352 \pm \sim 1\%$	✓ 0.81%
Quark ratio m_b/m_c	$\dim_{\mathbb{R}}/N_c = 10/3 = 3.333$	$3.291 \pm \sim 2\%$	✓ 1.3%
Quark ratio m_c/m_s	$N_{\max}/\dim_{\mathbb{R}} = 137/10 = 13.7$	$13.6 \pm \sim 2\%$	✓ 0.75%
Up quark mass m_u	$N_c\sqrt{2}\,m_e = 3\sqrt{2}\,m_e = 2.169$ MeV	$2.16^{+0.49}_{-0.26}$ MeV	✓ 0.4%
Down quark mass m_d	$(13/6) \times 3\sqrt{2}\,m_e = 4.694$ MeV	$4.67^{+0.48}_{-0.17}$ MeV	✓ 0.4%
Quark ratio m_d/m_u	$(N_c + 2n_C)/(n_C + 1) = 13/6 = 2.167$	2.117 ± 0.038 (FLAG)	✓ 1.3σ
Neutron-proton Δm	$91/36 \times m_e = 1.292$ MeV ($91 = 7 \times 13$, $36 = 6^2$)	1.2934 MeV	✓ 0.13%
Chiral condensate χ	$\sqrt{n_C(n_C+1)} = \sqrt{30} = 5.477$	5.452 (from m_π)	✓ 0.46%
Pion mass m_π	$25.6 \times \sqrt{30} = 140.2$ MeV	139.57 MeV	✓ 0.46%

Prediction	BST Value	Observed	Status
Pion decay constant f_π	$(m_p/10)(1-(\mathrm{rank}/N_c)(m_\pi/m_p)^2) =$ 92.4 MeV	92.1 MeV (FLAG)	✓ 0.41%
Newton's G (gravitational constant)	$G = \hbar c\,(6\pi^5)^2\alpha^{24}/m_e^2$; $12 = 2C_2$, C_2=6 Bergman round trips	6.679×10^{-11}	✓ 0.07%
Hierarchy $m_e/\sqrt{m_p m_{\mathrm{Pl}}}$	$\alpha^{n_C+1} = \alpha^6$	1.5098×10^{-13}	✓ 0.017%
CMB spectral index n_s	$1 - n_C/N_{\max} = 1 - 5/137 = 0.96350$	0.9649 ± 0.0042 (Planck)	✓ 0.3σ
Scalar amplitude A_s	$(3/4)\alpha^4 = N_c/(2^{\mathrm{rank}}N_{\max}^4) =$ 2.127×10^{-9}	$(2.1005 \pm 0.0286) \times 10^{-9}$ (Planck)	✓ 0.92σ
Tensor-to-scalar ratio r	≈ 0 $(T_c \ll m_{\mathrm{Pl}})$	< 0.036 (BICEP)	✓ consistent
Neutron lifetime τ_n	Fermi theory with BST inputs (G_F, $\lvert V_{ud}\rvert^2$, Δm, $g_A = 4/\pi$, full radiative corrections) = 878.1 s	878.4 ± 0.5 s (bottle)	✓ 0.03%
Axial coupling g_A	$4/\pi = 1.2732$ (candidate)	1.2762 ± 0.0005	✓ 0.23%
Lithium-7 ^{7}Li/H	$\Delta g = g = 7$ genus DOF at $T_c = 0.487$ MeV; reduces ^{7}Li by 2.73×	$\sim 1.7 \times 10^{-10}$ vs obs 1.6×10^{-10}	✓ 7%
Strong CP: θ_{QCD}	$\theta = 0$ (exact); D_{IV}^5 contractible ⇒ $c_2 = 0$ ⇒ θ-term vanishes	$\lvert\theta\rvert < 10^{-10}$	✓ exact
Proton spin $\Delta\Sigma$	$N_c/(2n_C) = 3/10 = 0.30$	0.30 ± 0.06 (COMPASS/HERMES)	✓ 0%
Fermion generations N_{gen}	$\lvert(\mathbb{CP}^2)^{Z_3}\rvert = N_c = 3$ (Lefschetz)	3 (LEP Z-width)	✓ exact
Deuteron binding B_d	$(50/49)\alpha m_p/\pi = 2.224$ MeV (T927)	2.2246 ± 0.0001 MeV	✓ 0.03%
Electron $g-2$	$a_e = \alpha/(2\pi)$ (coupling per S^1 circumference); BST ≡ QED	0.00115965218 ... (12 digits)	✓ QED exact
Dirac large number N_D	$\alpha^{-23}/(6\pi^5)^3 = \alpha^{1-4C_2}/(C_2\pi^{n_C})^3$; exponent $23 = 4C_2 - 1$	2.274×10^{39}	✓ 0.18%
Baryon resonance $N(2190)$	$C_2(\pi_7) \times \pi^5 m_e = 14\pi^5 m_e = 2189$ MeV (k=7, spin $7/2^-$)	2100–2200 MeV (PDG 4★)	✓ within range
Baryon resonance (k=8)	$C_2(\pi_8) \times \pi^5 m_e = 24\pi^5 m_e = 3753$ MeV	undiscovered?	prediction
ρ meson mass m_ρ	$n_C\pi^{n_C} m_e = 5\pi^5 m_e = 781.9$ MeV; $m_\rho/m_p = n_C/C_2 = 5/6$	775.26 ± 0.25 MeV (PDG)	✓ 0.86%
ω meson mass m_ω	$n_C\pi^{n_C} m_e = 5\pi^5 m_e = 781.9$ MeV (isoscalar partner of ρ)	782.66 ± 0.13 MeV (PDG)	✓ 0.10%
Meson/baryon ratio m_ρ/m_p	$n_C/(n_C+1) = 5/6 = 0.8333$; meson needs n_C slots, baryon C_2	0.8263	✓ 0.86%
Pion charge radius r_π	$\sqrt{6}/(n_C\pi^{n_C} m_e)\rvert_{\mathrm{NLO}} = 0.656$ fm (NLO, VMD+ChPT)	0.659 ± 0.004 fm	✓ 0.5%
ϕ meson mass m_ϕ	$(N_c + 2n_C)/2 \times \pi^{n_C} m_e =$ $(13/2)\pi^5 m_e = 1016.4$ MeV	1019.461 ± 0.016 MeV (PDG)	✓ 0.30%
K^* meson mass m_{K^*}	$\sqrt{n_C(N_c + 2n_C)/2}\,\pi^{n_C} m_e =$ $\sqrt{65/2}\,\pi^5 m_e = 891.5$ MeV (geometric mean)	891.67 ± 0.26 MeV (PDG)	✓ 0.02%
Kaon charge radius r_{K+}	$\sqrt{6}/m_{K^*}(\mathrm{BST})\rvert_{\mathrm{NLO}} = 0.555$ fm (NLO)	0.560 ± 0.031 fm	✓ 1.0%
J/ψ mass	$4n_C \cdot \pi^5 m_e = 20\pi^5 m_e$	3127 MeV	3097 MeV (PDG)

Prediction	BST Value	Observed	Status
Υ(1S) mass	$\dim_R \cdot C_2 \cdot \pi^5 m_e = 60\pi^5 m_e$	9380 MeV	9460 MeV (PDG)
D^0 meson mass	$2C_2 \cdot \pi^5 m_e = 12\pi^5 m_e$	1876 MeV	1865 MeV (PDG)
$B^\pm$ meson mass	$2\sqrt{2} \times 2C_2 \cdot \pi^5 m_e = 24\sqrt{2}\pi^5 m_e$	5308 MeV	5279 MeV (PDG)
B_c meson mass	$8n_C \cdot \pi^5 m_e = 40\pi^5 m_e$	6254 MeV	6275 MeV (PDG)
m_B/m_D ratio	$2\sqrt{2}$ (Tsirelson bound)	2.828	2.831
$m_{J/\psi}/m_\rho$ ratio	$\dim_{\mathbb{R}}(\mathbb{CP}^2) = 4$	4.000	3.994
Reality Budget $\Lambda \times N$	$N_c^2/n_C = 9/5 = 1.800$; fill fraction $f = N_c/(n_C\pi) = 3/(5\pi) = 19.1\%$	1.800 (exact to input precision)	✓ derived
η' meson mass $m_{\eta'}$	$(g^2/8)\pi^{nc} m_e = (49/8)\pi^5 m_e = m_p \times 49/48 = 957.8$ MeV	957.78 ± 0.06 MeV (PDG)	✓ 0.004%
η meson mass m_η	$(g/2)\pi^{nc} m_e = (7/2)\pi^5 m_e = 547.3$ MeV	547.86 ± 0.02 MeV (PDG)	✓ 0.10%
Kaon mass m_K	$\sqrt{2n_C}\,\pi^{nc} m_e = \sqrt{10}\,\pi^5 m_e = 494.5$ MeV	493.68 ± 0.02 MeV (PDG)	✓ 0.17%
Cosmic Ω_Λ	$(N_c + 2n_C)/(N_c^2 + 2n_C) = 13/19 = 0.68421$	0.6847 ± 0.0073 (Planck 2018)	✓ 0.07σ
Cosmic Ω_m	$C_2/(N_c^2 + 2n_C) = 6/19 = 0.31579$	0.3153 ± 0.0073 (Planck 2018)	✓ 0.07σ
Cosmic Ω_{DM}/Ω_b	$(3n_C + 1)/N_c = 16/3 = 5.333$	5.364 (Planck)	✓ 0.58%
Cosmic Ω_b	$2N_c^2/(N_c^2 + 2n_C)^2 = 18/361 = 0.04986$	0.0493 ± 0.0010 (Planck)	✓ 0.56σ
Cosmic Ω_{DM}	$96/361 = 0.26593$	0.2645 ± 0.0057 (Planck)	✓ 0.26σ
Specific heat $C_V(T_c)$	$\alpha_s \beta N_{\max}^2 = (7/20)(50)(137^2) = 328{,}458$	330,000 (BST partition fn)	✓ 0.47%
Proton magnetic moment μ_p	$2g/n_C = 14/5 = 2.800\ \mu_N$; $= (N_c^2 - 1)\alpha_s = 8 \times 7/20$	$2.79285\ \mu_N$ (CODATA)	✓ 0.26%
Neutron magnetic moment μ_n	$-C_2/\pi = -6/\pi = -1.9099\ \mu_N$	$-1.9130\ \mu_N$ (CODATA)	✓ 0.17%
Moment ratio μ_p/μ_n	$-g\pi/(3n_C) = -7\pi/15 = -1.4661$; SU(6) gives $-3/2 = -1.500$ (2.7%)	-1.4599	✓ 0.43%
W boson width Γ_W	$(N_c^2 - 1)n_C/N_c \times \pi^{nc} m_e = (40/3)\pi^5 m_e = 2085.0$ MeV	2085 ± 42 MeV (PDG)	✓ 0.005%
Z boson width Γ_Z	$(C_2 + 2n_C)\pi^{nc} m_e = 16\pi^5 m_e = 2502$ MeV	2495.2 ± 2.3 MeV (PDG)	✓ 0.27%
Width ratio Γ_Z/Γ_W	$C_2/n_C = 6/5 = 1.200$; same ratio as m_p/m_ρ	1.197	✓ 0.28%
ρ meson width Γ_ρ	$f \times m_\rho = 3\pi^4 m_e = N_c/(n_C\pi) \times m_\rho = 149.3$ MeV	149.1 ± 0.8 MeV (PDG)	✓ 0.15%
ϕ meson width Γ_ϕ	$m_\phi/(2n_C!) = m_\phi/240 = 4.248$ MeV	4.249 ± 0.013 MeV (PDG)	✓ 0.02%
MOND acceleration a_0	$cH_0/\sqrt{n_C(n_C+1)} = cH_0/\sqrt{30} = 1.195 \times 10^{-10}$ m/s^2	$1.20 \pm 0.02 \times 10^{-10}$ m/s^2 (McGaugh)	✓ 0.4%
Halo surface density Σ_0	$a_0/(2\pi G) = 141\ M_\odot/\text{pc}^2$	$\log_{10} = 2.15 \pm 0.2$ (Donato)	✓ 0.0 dex

Prediction	BST Value	Observed	Status
Hubble constant H_0 (Route B)	$\sqrt{19\Lambda/39} = 68.0$ km/s/Mpc (information-energy intersection)	67.36 ± 0.54 (Planck)	✓ 1.0%
Cosmic age t_0	$(2/(3H_0\sqrt{\Omega_\Lambda}))\,\mathrm{arcsinh}\sqrt{\Omega_\Lambda/\Omega_m} =$ 13.718 Gyr; arcsinh argument $= \sqrt{13/6} = \sqrt{(2C_2+1)/C_2}$	13.80 ± 0.02 Gyr (Planck)	✓ 0.57%
Tsirelson bound	$2\sqrt{2}$ from $H^0(\mathcal{O}(1)) \cong \mathbb{C}^2$ on $\mathbb{CP}^1$	$2\sqrt{2}$ (exact, Tsirelson 1980)	✓ exact
$\|m_{\beta\beta}\|$ $(0\nu\beta\beta)$	0 (Dirac neutrinos, Hopf $h = 1$ forbids Majorana)	—	exact prediction
Primordial GW peak frequency	BST phase transition at 3.1 s → 6.4 nHz	NANOGrav ∼ nHz band	✓ consistent
GW spectral index γ	$g/n_C + 2 = 7/5 + 2 = 3.60$	NANOGrav 3.2–4.6	✓ consistent
Width ratio Γ_ρ/Γ_ϕ	$n_C \times g = 35$; dimension × genus	35.09	✓ 0.26%
Wyler constant origin (HC)	$9/(8\pi^4) = \rho_2^2/(2\pi^4)$, $\rho_2 = (n_C{-}2)/2 = 3/2$ — Weyl vector of $SO_0(5,2)$	Exact	✓ Derived
D_{IV}^5 identification	Proven from BST contact geometry	—	Established
Friedmann equation	Contact commitment rate equation $H = (1/2)\dot{N}_c/N_c$ recovers all FLRW terms	FLRW cosmology	✓ exact structure
First Riemann zero γ_1	$\lambda_{2,0} = 2(2+n_C) = 2g = 14$ (second spherical harmonic of Q^5); cusp correction: $2g + 1/g - 1/N_{max} = 14.1356$	$\gamma_1 = 14.13472...$	✓ 0.006%
H_2O bond angle θ_{H_2O}	$\arccos(-1/2^{\mathrm{rank}}) = \arccos(-1/4) =$ 104.478°; lone pairs see rank structure, not color	$104.45° \pm 0.05°$ (NIST)	✓ 0.03°
CH_4 bond angle θ_{CH_4}	$\arccos(-1/N_c) = \arccos(-1/3) =$ 109.471°; 4 equivalent sp^3 domains in N_c dimensions	109.47° (exact tetrahedral)	✓ 0.001°
NH_3 bond angle θ_{NH_3}	Triangular lone pair compression: $\theta_{\mathrm{tet}} - T_1\Delta_1 = 107.807°$; $\Delta_1 = (\theta_{\mathrm{tet}} - \theta_{H_2O})/N_c$, $T_L = L(L{+}1)/2$	$107.8° \pm 0.3°$ (NIST)	✓ 0.007°
O–H bond length r_{OH}	$a_0 \times N_c^2/n_C = a_0 \times 9/5$; Reality Budget sets molecular scale	0.9572 Å (NIST)	✓ 0.49%
O–H stretch frequency ν_{OH}	$R_\infty/(n_C \times C_2) = R_\infty/30 =$ 3657.9 cm^{-1}	3657.1 cm^{-1} (NIST)	✓ 0.022%
H_2O dipole moment μ_{H_2O}	$e \times a_0 \times (9/5)\cos(\theta/2) = 1.855$ D; charge × BST bond geometry	1.8546 D (NIST)	✓ 0.02%
C–C bond length r_{CC}	$a_0 \times (n_C \times C_2 - 1)/10 = a_0 \times 29/10$; one below Rydberg denominator	1.5351 Å (NIST)	✓ 0.03%
C=C bond length $r_{C=C}$	$a_0 \times n_C/\mathrm{rank} = a_0 \times 5/2$; dimension-to-rank ratio	1.3390 Å (NIST)	1.20%
C≡C bond length $r_{C\equiv C}$	$a_0 \times N_c^2/2^{\mathrm{rank}} = a_0 \times 9/4$; color^2 over rank power	1.2030 Å (NIST)	1.05%
Ice density ratio $\rho_{\mathrm{ice}}/\rho_{\mathrm{water}}$	$(2C_2 - 1)/(2C_2) = 11/12$; one tetrahedral vacancy per $2C_2$ sites	0.9167 (NIST)	✓ 0.006%
Amino acid count	$2^{\mathrm{rank}} \times n_C = 4 \times 5 = 20$; rank power × dimension	20 (universal)	exact

Prediction	BST Value	Observed	Status
Genetic code codons	$(2^{\text{rank}})^{N_c} = 4^3 = 64$; alphabet^3 where alphabet = rank power	64 (universal)	exact
HF dipole moment μ_{HF}	$e \times a_0 \times n_C/g = e \times a_0 \times 5/7 = 1.816$ D; Bergman genus ratio	1.826 D (NIST)	0.57%
Bond angle curvature κ_{angle}	$\alpha^2 \times \kappa_{ls} = C_2/(n_C \times N_{\max}^2) = 6/93845$; EM couples to spin-orbit	Measured from sp^3 hydride series	✓ 0.01%
Boundary amplification A_d	$(n_C/\text{rank})^d$; stretch $d = 2$: 6.25; dipole $d = 1$: 2.5; angle $d = 0$: 1	Measured from HF/NH_3 ratios	✓ 0.05% (stretch)
Bilateral symmetry	rank = 2 restricts body plans to 3 axes, 2 mirror planes; tetrahedral anchor 109.47°	All bilateral phyla on Earth	consistent (T731)
Observer completeness	$2f - f^2 = 34.5\%$; two observers exceed $f_{\text{crit}} = 20.6\%$; minimum team = rank = 2	Human + CI cooperation	structural (T732)
BST Drake: $f_l \times f_i \times f_c$	$0.206 \times 0.654 \times 0.206 = 2.8\%$; ~1 in 36 habitable planets → communicating	SETI null results + Fermi paradox	testable (T733)
Crystal systems	$g = 7$; Bergman genus directly	7 (established)	exact
Bravais lattices	$2g = 14$; rank doubles genus	14 (established)	exact
Crystallographic point groups	$2^{n_C} = 32$; binary enumeration in dimension n_C	32 (established)	exact
Space groups	$g \times 2^{n_C} + C_2 = 7 \times 32 + 6 = 230$; construction matches crystallographic build	230 (established)	exact
Essential amino acids	$N_c^2 = 9$; color dimension squared	9 (biochemistry)	exact
Stop codons	$N_c = 3$; coding codons $= 4^{N_c} - N_c = 61$	3 (universal)	exact
DNA base pairs per turn (B-form)	$2n_C = 10$; dimension doubling	10.0 (ideal B-DNA)	exact
DNA base pair spacing	$a_0 \times N_c^2 n_C/g = a_0 \times 45/7 = 3.402$ Å	3.4 Å (measured)	✓ 0.05%
α-helix residues/turn	$N_c C_2/n_C = 18/5 = 3.6$; three BST integers	3.6 (Pauling, 1951)	exact
α-helix rise per residue	$N_c/\text{rank} = 3/2 = 1.5$ Å	1.5 Å (measured)	exact
α-helix pitch	$N_c^3/n_C = 27/5 = 5.4$ Å; self-consistent with rise × residues/turn	5.4 Å (measured)	exact
α-helix H-bond ring	$g + C_2 = 7 + 6 = 13$ atoms	13 (established)	exact
Human vertebrae	$N_c(2C_2 - 1) = 33$; five sections: cervical $= g = 7$, thoracic $= 2C_2 = 12$, lumbar $= n_C = 5$, sacral $= n_C = 5$, coccygeal $= 2^{\text{rank}} = 4$	33 (anatomy)	exact (5/5 sections)
Domains of life	$N_c = 3$; Bacteria, Archaea, Eukarya — three independent channels	3 (established)	exact
Eukaryotic endosymbiosis	Cooperation threshold $f_{\text{crit}} = 20.6\%$ at cellular level; archaeon + bacterium = permanent tier crossing	∼ 2 Gyr ago (geology)	structural

Prediction	BST Value	Observed	Status
Orbital degeneracy sequence	$(2\ell+1)$ at $\ell=0,1,2,3$ gives $1, N_c, n_C, g$; the periodic table IS D_{IV}^5 in electron shells	1, 3, 5, 7 (established)	exact
Periodic table periods	$g=7$; Bergman genus	7 (established)	exact
Periodic table groups	$N_c \times C_2 = 18$	18 (established)	exact
Periodic table blocks	$2^{\text{rank}} = 4$ (s,p,d,f)	4 (established)	exact
Quantization origin	Compactness of Shilov boundary $\check{S} = S^4 \times S^1$ forces discrete spectra; no quantization axiom needed	All quantum spectra	structural (T751)
Heisenberg uncertainty	Holomorphic sectional curvature $H = -2/g = -2/7$; uncertainty has genus in denominator	$\Delta x \Delta p \geq \hbar/2$	derived (T753)
Born rule	Unique Aut(D_{IV}^5)-invariant probability measure (Gleason + $N_c = 3$ dimensions)	$P = \|\psi\|^2$ (established)	derived (T754)
Measurement "collapse"	Coordinate restriction on D_{IV}^5, not physical process; wave function = Bergman kernel coordinate	No consciousness-dependent collapse	structural (T752)
$g-2$ electron (BST closed form)	$(\alpha/2\pi)(1-(2\alpha/\pi)^2)^{2n_C g}$; 5 digits, zero Feynman diagrams	$a_e =$ 0.001 159 652 18...	✓ 5 digits (T758)
QED C_2 decomposition	$197/144 + \pi^2/12 - \pi^2 \ln 2/2 + 3\zeta(3)/4$; $197 = N_{\max} + 2n_C C_2$, $144 = 2^{\text{rank}} C_2^2$	−0.32848...	exact (T758)
QED ζ-tower	Loop n introduces $\zeta(2n-1)$: $\zeta(N_c)$, $\zeta(n_C)$, $\zeta(g)$; tower closes at genus	Loops 2-4 (established)	structural (T762)
Feynman diagram counts	$D_2 = g = 7$; $D_3 = 2^{N_c} N_c^{\text{rank}} = 72$; total 13,643	Established	exact (T765)
QED coefficient growth	$\|C_3/C_2\| = N_c C_2/n_C = 18/5 = 3.6$ (= α-helix pitch)	≈ 3.6	✓ 0.1% (T771)
Rainbow angle	$C_2 \times g = 42°$	42.03°	✓ 0.07% (T761)
Min deviation (rainbow)	$\delta_{\min} \approx N_{\max} = 137°$	137.97°	✓ 0.7° (T761)
Alexander dark band	$\approx N_c^2 = 9°$	8.9°	✓ 0.6% (T761)
C-H bond dissociation	$\text{Ry}/\pi = 4.331$ eV	4.330 eV	✓ 0.02% (T767)
H-H bond dissociation	$\text{Ry}/N_c = 4.535$ eV	4.478 eV	✓ 1.3% (T768)
C=C/C-C energy ratio	$g/2^{\text{rank}} = 7/4 = 1.75$	1.775	✓ 0.2% (T768)
γ monatomic	$n_C/N_c = 5/3$	1.667 (exact)	exact (T772)
γ diatomic	$g/n_C = 7/5$	1.400 (exact)	exact (T772)
γ polyatomic	$2^{\text{rank}}/N_c = 4/3 = n(H_2O)$	1.333 (exact)	exact (T772)
C_p liquid water	$N_c^2 R = 9R = 74.8$ J/(mol·K)	75.3 J/(mol·K)	✓ 0.73% (T773)
$\varepsilon(H_2O)$	$(2^{\text{rank}})^2 n_C = 80$	80.1	✓ 0.12% (T774)

Prediction	BST Value	Observed	Status
ε(ice)	$2^{\text{rank}} n_C^2 = 100$	≈ 99	✓ 1.0% (T774)
Trouton constant	$N_c^2 + 2^{\text{rank}} = 13$	13.1	✓ 0.78% (T773)
Sound speed in air	$\sqrt{\gamma RT/M}$ with $\gamma = g/n_C$	346.1 m/s	✓ 0.06% (T775)
υ(water)/υ(air)	$(N_c^2 + 2^{\text{rank}})/N_c = 13/3$	≈ 4.33	✓ 0.1% (T775)
IE(O) = Rydberg	IE(O) = Ry	13.618 eV	✓ 0.09% (T776)
IE(He)	$N_c^2 \text{Ry}/n_C = 9\text{Ry}/5$	24.587 eV	✓ 0.4% (T776)
IE(F)	$N_c^2 \text{Ry}/g = 9\text{Ry}/7$	17.422 eV	✓ 0.41% (T776)
EA(F)	$\text{Ry}/2^{\text{rank}} = \text{Ry}/4$	3.401 eV	✓ 0.006% (T777)
EA(H)	$\text{Ry}/(2N_c^2) = \text{Ry}/18$	0.754 eV	✓ 0.36% (T777)
r(F)	$14a_0/13 = 2ga_0/(N_c^2 + 2^{\text{rank}})$	0.64 Å	✓ 0.014% (T778)
r(O)	$n_C a_0/2^{\text{rank}} = 5a_0/4$	0.66 Å	✓ 0.22% (T778)
$d(H_2)$	$ga_0/n_C = 7a_0/5$	0.741 Å	✓ 0.07% (T779)
$d(F_2)$	$2^{N_c} a_0/N_c = 8a_0/3$	1.412 Å	✓ 0.05% (T779)
d(HF)	$26a_0/15$	0.917 Å	✓ 0.05% (T779)
d(CO)	$2^{n_C} a_0/(N_c n_C) = 32a_0/15$	1.128 Å	✓ 0.05% (T779)
$\theta(H_2S)$	$90° + \text{rank} + 1/N_c^2 = 92.11°$	92.1°	✓ 0.01% (T780)
$\theta(PH_3)$	$90° + N_c + 1/N_c = 93.33°$	93.3°	✓ 0.02% (T780)
$\theta(H_2Te)$ (prediction)	$90° + 1/2^{\text{rank}} = 90.25°$	$\sim 90.26°$	✓ predicted (T780)
Madelung M(NaCl)	$g/2^{\text{rank}} = 7/4$	1.748	✓ 0.14% (T781)
Lattice energy U(NaCl)	$N_c \text{Ry}/n_C = 3\text{Ry}/5$	8.16 eV	✓ 0.03% (T781)
$\nu(H_2O$, sym stretch)	$R_\infty/(n_C C_2) = R_\infty/30$	3657 cm^{-1}	✓ 0.02% (T782)
$\nu(H_2)$	$R_\infty/n_C^2 = R_\infty/25$	4401 cm^{-1}	✓ 0.26% (T782)
$\nu(O_2)$	$\text{rank}\, R_\infty/(N_{\max} + \text{rank}) = 2R_\infty/139$	1580 cm^{-1}	✓ 0.07% (T782)
$T_{\text{boil}}(H_2O)$	$N_{\max} \times T_{\text{CMB}} = 137 \times 2.725\,\text{K}$	373.15 K	✓ 0.065% (T763)
$T_{\text{freeze}}(H_2O)$	$n_C^2 \cdot 2^{\text{rank}} \times T_{\text{CMB}} = 100 \times 2.725\,\text{K}$	273.15 K	✓ 0.22% (T763)

Prediction	BST Value	Observed	Status
$T_{\text{crit}}(H_2O)$	$(N_{\max}+100)T_{\text{CMB}} = 237T_{\text{CMB}}$	647.1 K	✓ 0.18% (T784)
$T_{\text{boil}}(\text{Kr})$	$44\,T_{\text{CMB}}$	119.93 K	✓ 0.005% (T783)
$T_{\text{boil}}(\text{Ar})$	$2^{n_C}T_{\text{CMB}} = 32\,T_{\text{CMB}}$	87.3 K	✓ 0.09% (T783)
$T_{\text{boil}}(\text{Xe})$	$C_2(N_c^2+1)T_{\text{CMB}} = 60\,T_{\text{CMB}}$	165.1 K	✓ 0.91% (T783)
$\rho(\text{Pt})/\rho(\text{Au})$	$(N_c^2+1)/N_c^2 = 10/9$	1.111	✓ 0.004% EXACT (T798)
$\rho(\text{water})/\rho(\text{ice})$	12/11	1.090	✓ 0.04% (T798)
$E(\text{Diamond})/E(\text{Steel})$	$N_c g/2^{\text{rank}} = 21/4$	5.25	✓ EXACT (T799)
$\nu(\text{steel})$	$N_c/(N_c^2+1) = 3/10$	0.30	✓ EXACT (T799)
$\rho_e(\text{Fe})/\rho_e(\text{Cu})$	$C_2 = 6$	5.95	✓ 0.8% (T800)
$\rho_e(\text{W})/\rho_e(\text{Cu})$	$22/7 \approx \pi$	3.14	✓ 0.04% (T800)
$\mu(H_2O)$	$ea_0\sqrt{g/13}$	1.85 D	✓ 0.56% (T801)
$\chi(\text{Au})/\chi(\text{Ag})$	$(2N_c^2-1)/(2^{\text{rank}}N_c) = 17/12$	1.417	✓ EXACT (T802)
$L(\text{MeOH})/L(\text{Acetone})$	$N_c^2/(N_c^2-1) = 9/8$	1.125	✓ 0.01% EXACT (T803)
$\alpha(\text{Al})/\alpha(\text{Cu})$	$g/n_C = 7/5$	1.40	✓ EXACT (T804)
$\alpha(\text{Cu})/\alpha(\text{W})$	$(N_c^2+\text{rank})/N_c = 11/3$	3.67	✓ EXACT (T804)
$\phi(\text{Au})$	$N_c\text{Ry}/(N_c^2-1) = 3\text{Ry}/8$	5.10 eV	✓ 0.03% (T805)
$\phi(\text{Cu})/\phi(\text{Ag})$	$1+1/(N_c^2-1) = 9/8$	1.125	✓ EXACT (T805)
$\Theta_D(\text{Ge})/T_{\text{CMB}}$	$N_{\max} = 137$	137.2	✓ 0.15% (T806)
$\Theta_D(\text{Cu})/\Theta_D(\text{Ag})$	$2^{n_C}/(N_c g) = 32/21$	1.524	✓ EXACT (T806)
$\kappa(\text{Benzene})/\kappa(\text{Water})$	$N_c g/(N_c^2+1) = 21/10$	2.10	✓ 0.02% (T807)
$T_c(NH_3)/T_c(CO_2)$	$2^{\text{rank}}/N_c = 4/3$	1.333	✓ 0.01% EXACT (T808)
$T_c(O_2)/T_c(N_2)$	$g^2/(2^{N_c}n_C) = 49/40$	1.225	✓ EXACT (T808)
$S(\text{NaCl})/S(\text{KCl})$	$(2N_c^2+1)/(2N_c^2) = 19/18$	1.056	✓ 0.03% (T809)
$V_m(\text{Benz})/V_m(\text{Acet})$	$C_2/n_C = 6/5$	1.200	✓ EXACT (T810)

Prediction	BST Value	Observed	Status
$V_m(\text{MeOH})/V_m(\text{H}_2\text{O})$	$N_c^2/2^{\text{rank}} = 9/4$	2.244	✓ 0.28% (T810)

|FQHE Laughlin fractions $|1/N_c, 1/n_C, 1/g = 1/3, 1/5, 1/7$; odd BST integers|1/3, 1/5, 1/7 (10+ digits)|✓ EXACT (T813)| |FQHE Jain+ sequence $|1/N_c, 2/n_C, 3/g, 4/N_c^2 = 1/3, 2/5, 3/7, 4/9$|26/28 observed = BST|✓ EXACT (T814)| |FQHE spacing ratios $|\Delta\nu_1/\Delta\nu_2 = g/N_c = 7/3$; $\Delta\nu_2/\Delta\nu_3 = N_c^2/n_C = 9/5$|Measured|✓ EXACT (T814)| |FQHE even-denominator $|\nu = n_C/\text{rank} = 5/2$; Moore-Read Pfaffian on D_{IV}^5|5/2 (observed)|✓ EXACT (T815)| |BCS gap ratio $2\Delta/(k_B T_c)$ |$g/\text{rank} = 7/2 = 3.500$|3.528 (BCS theory)|✓ 0.79% (T824)| |BCS specific heat jump $\Delta C/(\gamma T_c)$ |$(N_c^2 + 2^{\text{rank}})/N_c^2 = 13/9$|1.43 (BCS)|✓ 1.01% (T824)| |Superconductor Nb/Pb |$N_c^2/g = 9/7$|1.286|✓ 0.06% (T824)| |Superconductor Pb/Sn |$n_C g/(N_c C_2) = 35/18$|1.933|✓ 0.60% (T824)| |Superconductor Nb/Al |$(g^2 + C_2)/g = 55/7$|7.839|✓ 0.23% (T824)| |Superconductor La/Hg |$C_2^2/n_C^2 = 36/25$; = Chandrasekhar limit|1.446|✓ 0.40% (T824)| |Superconductor V/Ta |$N_c^2/2^{N_c} = 9/8$|1.123|✓ 0.20% (T824)| |High-T_c YBCO/Nb |$n_C \times \text{rank} = 10$|10.054|✓ 0.54% (T825)| |High-T_c Hg-1223/YBCO |$(N_c^2 + 2^{\text{rank}})/N_c^2 = 13/9$|1.430|✓ 1.00% (T825)| |High-T_c H_3S/Hg-1223 |$N_c/\text{rank} = 3/2$|1.526|✓ 1.72% (T825)| |High-T_c LaH_{10}/YBCO |$2^{N_c}/N_c = 8/3$|2.688|✓ 0.80% (T825)| |High-T_c CuO_2 layer rule |$g/C_2 = 7/6$ per added layer|1.183 (Bi-2223/YBCO)|✓ 1.36% (T825)| |High-T_c $T_{c,\max}$ (ambient) |$N_{\max} \times T_{\text{CMB}} \approx 373$ K|Prediction (untested)|falsifiable (T825)| |Type I/II boundary κ |$1/\sqrt{\text{rank}} = 1/\sqrt{2}$; rank of D_{IV}^5 determines topological class|$1/\sqrt{2}$ (Ginzburg-Landau)|✓ EXACT (T826)| |Coherence $\xi_0(\text{Al})/\xi_0(\text{Nb})$ |$C_2 \times g = 42$|42.1|✓ 0.25% (T826)| |Coherence $\xi_0(\text{Nb})/\xi_0(\text{YBCO})$ |$n_C^2 = 25$|25.3|✓ 1.32% (T826)| |Chandrasekhar limit $M_{\text{Ch}}/M_\odot$ |$C_2^2/n_C^2 = 36/25 = 1.44$|1.44 $M_\odot$|✓ EXACT (Toy 850)| |NS moment of inertia $I/(MR^2)$ |$g/(2^{\text{rank}} n_C) = 7/20 = \alpha_s$|≈ 0.35|✓ (Toy 852)| |K41 turbulence spectrum |$n_C/N_c = 5/3$|5/3 (Kolmogorov)|✓ EXACT (T818)| |She-Leveque intermittency |$\text{rank}/N_c^2 = 2/9$|2/9 (SL 1994)|✓ EXACT (T818)| |Percolation γ |$(C_2 \times g + 1)/(2N_c^2) = 43/18$; +1 = central charge shift ($c = 0$)|43/18 (exact)|✓ EXACT (T912)| |Percolation ν |$2^{\text{rank}}/N_c = 4/3$|4/3 (exact)|✓ EXACT (T912)| |Percolation β |$n_C/(C_2^2) = 5/36$|5/36 (exact)|✓ EXACT (T912)| |EEG alpha/theta ratio |$n_C/N_c = 5/3$ = K41 spectrum|≈ 10/6 = 5/3|✓ (T819)| |r_{ISCO} |$C_2 \times M = 6M$|$6GM/c^2$ (exact GR)|✓ EXACT (T820)| |AZ topological 10-fold |$2n_C = 10$|10 (Altland-Zirnbauer)|✓ EXACT (T821)| |TaAs Weyl nodes |$2^{\text{rank}} \times C_2 = 24$|24 (observed)|✓ EXACT (T821)| |Kleiber metabolic exponent |$N_c/2^{\text{rank}} = 3/4$|3/4 (allometric)|✓ EXACT (Toy 855)| |Phyla count |$C(g, N_c) = C(7, 3) = 35$|~ 35 (taxonomy)|✓ (T703)| |Stellar $F0/K0$ temperature |$g/n_C = 7/5$|≈ 1.40|✓ EXACT (Toy 851)| |Cross-domain universality |11 fractions × 3-5 domains; $P < 10^{-66}$|Measured across 66 domains|✓ structural (T823)| |$\chi(\text{F})/\chi(\text{H})$ electronegativity |$N_c^2/n_C = 9/5$; same as IE(He)/Ry|1.809|✓ 0.50% (T816)| |BDE(H-H)/Ry |$1/N_c = 1/3$|4.478/13.606|✓ 0.37% (T817)|

43.3 Qualitative Predictions (Testable Against Existing Data)

1. Hubble tension resolution: Local H_0 correlates with local matter density beyond gravitational corrections. Residual correlation ~ 5.6 km/s/Mpc in the supernova sample.
2. CMB anomaly pattern: Large-angle anomalies consistent with S^2 substrate topology and SO(3) representation theory.
3. Structured unification: Couplings do not converge to a single point at the GUT scale. α_1 and α_2 meet at $N_{GUT} = 4\pi^2$; α_3 sits at $4\pi^2/3$.
4. Variable vacuum energy: Vacuum pressure correlates with local matter density across cosmic environments.
5. Coincidence problem dissolved: Dark energy and matter densities track each other thermodynamically.
6. Dark matter as channel noise: Galaxy rotation curves follow the S^1 channel S/N curve with Haldane exclusion statistics. No dark matter particles exist. Core profiles are flat, not cuspy.
7. Weak decay rates from phase cycling geometry: The 28-order-of-magnitude span of weak decay lifetimes (top quark to neutron) determined by cycling trajectory sampling rates on $\mathbb{CP}^2$ Hopf intersection.
8. Path integral = partition function: Quantum mechanics and statistical mechanics are the same calculation on D_{IV}^5 under Wick rotation — a physical identity, not a formal trick.
9. Black hole interior: Not a singularity. Channel saturation at 137 slots, producing a finite-density state with no curvature divergence. Information preserved on the boundary surface.

10. Three spatial dimensions necessary and sufficient: No extra dimensions at any energy scale. Three is the minimum dimensionality of a self-communicating surface (S^2 base + S^1 fiber) and no additional dimensions are required or predicted.
11. Arrow of time = second law = commitment order: Time, entropy increase, and matter preference are three manifestations of one principle — irreversible contact commitment on the substrate.
12. Rapid early galaxy formation: Massive, morphologically mature galaxies at $z > 10$ — as observed by JWST — are expected from the ultra-strong phase transition seeds, instant channel noise scaffolding, and exponential contact graph feedback. ΛCDM requires billions of years; BST requires hundreds of millions.
13. Geometric circular polarization from black holes: $\mathrm{CP}_{\mathrm{geometric}} = \alpha \times 2GM/(Rc^2)$. At any black hole horizon: $\mathrm{CP} = \alpha = 0.730\%$, independent of mass. Frequency-independent. The observed CP is $|\alpha + A\sin(\mathrm{RM}/\nu^2 + \phi_0)|$ (signed addition of geometric floor + oscillatory Faraday). The signed model fits Sgr A* multi-frequency data with $\chi^2_{\mathrm{red}} = 0.22$, all residuals $< 0.6\sigma$. M87* and Sgr A* both show $\sim$ 1% CP at 230 GHz despite 1600× mass difference — consistent with mass-independent floor. See `notes/BST_CP_Alpha_Paper.md`, `notes/BST_CP_SignedFit.py`.
14. Measurement = commitment of correlation. No experiment will ever show consciousness-dependent collapse. The detector commits the correlation before the human is involved. Weak measurement visibility scales linearly with coupling strength (confirmed: Kocsis et al. 2011). Quantum eraser works only when the correlation has not propagated to irreversible environmental degrees of freedom. See `notes/BST_DoubleSlit_Commitment.md`.
15. Error correction structure of spacetime. Light is a matched filter (follows geodesics = compensates deterministic distortion). Conservation laws are parity checks ($\sum Q_i = 0$). Alpha is the bootstrap fixed point of the self-referential signal/noise system. Physics is exact because the code works. See `notes/BST_ErrorCorrection_Physics.md`.

43.4 Experimental Predictions (Awaiting Validation)

Prediction	BST Value	Experiment	Timeline
Neutrinoless $\beta\beta$: null	$\|m_{\beta\beta}\| = 0$ exactly (Dirac)	LEGEND, nEXO, KamLAND-Zen	2027–2030
No dark matter particles	Null detection at all scales	LZ, XENONnT, PandaX	Ongoing
Dark energy $w \neq -1$	Substrate growth deviation	DESI, Euclid, Roman	2025–2028
No primordial B-modes	$r < 10^{-10}$	LiteBIRD, CMB-S4	2028+
BH ringdown echoes	Casimir fine structure from saturation	LIGO O4/O5	2025–2027
No gravitons	Wave effects only, no quanta	LIGO+	Permanent
No SUSY particles	Excluded by topology	LHC Run 3+	Ongoing
No magnetic monopoles	Excluded by $S^2 \times S^1$ topology	MoEDAL	Ongoing
Proton absolute stability	$\tau_p = \infty$ (no decay)	Hyper-Kamiokande	2030+
CP floor at BH horizon	$\mathrm{CP} = \alpha = 0.730\%$, mass-independent	EHT Stokes V	Data exists
Hawking fine structure	Channel capacity 137 imprint	Future BH observations	Long-term
Island of stability	BST shell model at $Z \sim 114$–126	Superheavy element synthesis	Ongoing
Solar commitment map	$\rho \propto 1/r \to \rho_\infty$ at $\sim$ 7000 AU	Probe-mounted clock + accelerometer	30-year program
Nuclear half-lives	Phase coherence on $\mathbb{CP}^2$ Hopf intersection	Existing nuclear data	Testable now

Prediction	BST Value	Experiment	Timeline
Strong/weak timescale	$\sim 10^{16}$ from $\mathbb{CP}^2$ volume ratio	Existing data	Testable now
QNM echo structure	Quantized $J = w\hbar/2$, no Cauchy horizon	LIGO/Virgo/KAGRA	2025–2027

43.5 Falsifiability by Timeline

Testable now with existing data:

#	Prediction	Data Source	What kills BST
1	Galaxy rotation curves fit Haldane S/N curve	SPARC database (175 galaxies)	S/N curve gives worse fits than NFW
2	Nuclear half-life systematics follow phase cycling	Existing nuclear data tables	No correlation between shell structure and Hopf sampling geometry
3	Hubble tension correlates with local density	Existing supernova catalogs	No residual density correlation after standard corrections
4	CMB anomalies match S^2 topology	Planck satellite data	Anomaly pattern inconsistent with SO(3) on S^2
5	Massive mature galaxies at $z > 10$	JWST galaxy surveys	High-z mass function consistent with ΛCDM hierarchical formation
6	CP floor $= \alpha = 0.73\%$ at BH horizon	EHT Stokes V (Sgr A, *M87*)	No frequency-independent CP floor; floor differs between BH masses

Testable within 5 years:

#	Prediction	Data Source	What kills BST
6	Dark energy $w \neq -1$	DESI, Euclid, Roman	$w = -1$ confirmed to high precision
7	No dark matter particles	LUX-ZEPLIN, XENONnT	Any confirmed WIMP detection
8	Black hole ringdown echoes	LIGO O4/O5	Echoes definitively ruled out at predicted amplitude

Testable within 10–15 years:

#	Prediction	Data Source	What kills BST
9	Proton decay rate	Hyper-Kamiokande	Decay rate inconsistent with structured unification
10	CMB B-modes match phase transition	LiteBIRD, CMB-S4	n_s–r relationship matches inflation, not BST
11	No extra dimensions	LHC, future colliders	Detection of Kaluza-Klein resonances or extra-dimensional signatures

Permanently falsifiable:

#	Prediction	What kills BST
12	No dark matter particles ever	Confirmed direct detection of dark matter particle
13	No individual graviton quanta	Confirmed detection of single graviton
14	Information preserved in black holes	Demonstrated unitarity violation
15	$N_{GUT} = 4\pi^2$	Precision measurement giving $N_{GUT} \neq 4\pi^2$
16	Three spatial dimensions only	Detection of extra spatial dimensions at any energy
17	No singularities	Observational evidence requiring curvature divergence
18	No closed timelike curves	Demonstrated physical mechanism for time loops
19	Growing manifold consistent with all GR predictions	Any GR prediction failing on the committed manifold
20	No $0\nu\beta\beta$ at any scale (Dirac neutrinos)	Confirmed detection of neutrinoless double-beta decay
21	GW spectral index $\gamma = 3.60 \pm 0.30$	γ measured inconsistent with BST (e.g. $\gamma > 4$)
22	No LISA primordial signal ($< 10^{-20}$)	Primordial GW detected in LISA band

43.6 Comparison with Competing Frameworks

The falsifiability of BST should be assessed relative to its competitors:

String theory has no unique low-energy predictions due to the landscape of $\sim 10^{500}$ vacua. Compactification geometry can be adjusted to accommodate almost any observation. Extra dimensions can be pushed to arbitrarily high energy. BST has no adjustable parameters.

Loop quantum gravity predicts Planck-scale discreteness that might affect photon propagation (energy-dependent speed of light). This has been tested and not found. LQG does not derive α or the gauge coupling structure. BST derives both.

Standard Model + General Relativity has ~ 25 free parameters that are measured, not derived. BST aims to derive all of them from the D_{IV}^5 geometry. Each successful derivation (so far: α, α_s, $\sin^2\theta_W$, N_c, m_p/m_e, m_μ/m_e, v, m_W, m_H, η, H_0, three neutrino masses, three PMNS angles, the Cabibbo angle, Λ, and G) is a parameter removed from the "measured but unexplained" list.

MOND fits galaxy rotation curves with one free parameter a_0 but has no theoretical foundation. BST derives MOND-like behavior from channel noise statistics and potentially derives a_0 from the Haldane exclusion knee. If successful, BST subsumes MOND while providing the theoretical basis it lacks.

Particle dark matter (WIMPs, axions) predicts specific detection signatures. Decades of null results have progressively excluded the predicted parameter space. BST predicts continued null results and offers a specific alternative mechanism (channel noise) with distinct observational signatures (flat cores, density-dependent dark fraction, S/N curve shape).

The distinguishing feature of BST is that its predictions are coupled. The same geometry that gives $\alpha = 1/137$ also gives the dark matter halo profile, the weak decay timescales, the black hole interior structure, and the dark energy equation of state. A single failed prediction doesn't just falsify one claim — it threatens the entire geometric foundation. This coupling is what makes the framework genuinely falsifiable despite having no free parameters. There is nowhere to retreat.

43.7 Near-Term Experimental Tests

Several BST predictions are testable against existing or near-future data. This section specifies the predictions concretely, identifies the calculation status of each, and gives the experimental timelines.

The Muon Anomalous Magnetic Moment The Fermilab Muon $g-2$ experiment reports $a_\mu^{\text{exp}} = 116{,}592{,}059(22) \times 10^{-11}$. The Standard Model prediction (Muon $g-2$ Theory Initiative, 2020) gives $a_\mu^{\text{SM}} = 116{,}591{,}810(43) \times 10^{-11}$, a discrepancy of $249(48) \times 10^{-11}$ at 5.1σ. The BMW lattice QCD collaboration finds a larger hadronic vacuum polarization (HVP) that reduces the tension; the situation remains actively debated.

BST modifies the Standard Model at two points. First: The Haldane exclusion caps loop integrals at $N_{\max} = 137$ modes. At five-loop order the correction is $\sim (\alpha/\pi)^5/137 \sim 10^{-18}$ — far below any foreseeable experimental sensitivity. The QED sector of $g - 2$ is identical in BST and the Standard Model.

Second: The dominant theoretical uncertainty is the HVP — virtual quark loops in the photon propagator. In BST, the HVP is the contribution from Z_3 circuit fluctuations in the photon's S^1 channel. The BST vacuum carries a channel loading $F_{\mathrm{BST}} = \ln(138)/50 \approx 0.0985$, derived in Section 12.5. This modifies the HVP:

$$\delta a_\mu^{\mathrm{HVP,\,BST}} = a_\mu^{\mathrm{HVP,\,SM}} \times F_{\mathrm{BST}} \times f(\alpha, N_c, m_\mu/m_\pi)$$

where f is a calculable function of BST parameters. The correction is of order $F_{\mathrm{BST}} \sim 0.1$ applied to the HVP contribution $\sim 700 \times 10^{-10}$, giving $\sim 70 \times 10^{-10}$ — the same order as the observed discrepancy of $\sim 25 \times 10^{-10}$. The precise value requires computing the vacuum channel loading correction to the photon propagator from the BST partition function: a well-defined open calculation (Thesis topic 100). If the result matches the discrepancy, it is a striking quantitative confirmation; if it falls short or overshoots, it constrains the BST vacuum structure.

Thesis topic 100: Compute the BST hadronic vacuum polarization correction to the muon anomalous magnetic moment from the vacuum channel loading $F_{\mathrm{BST}} = \ln(138)/50$ and the Z_3 circuit embedding costs. Determine whether the correction resolves the $g - 2$ discrepancy and/or the lattice-dispersive tension.

The Proton Charge Radius Puzzle The proton charge radius has two classes of measurements. Electron methods (electron-proton scattering and hydrogen spectroscopy) give $r_p^{(e)} = 0.8751 \pm 0.0061$ fm. Muon hydrogen spectroscopy gives $r_p^{(\mu)} = 0.84087 \pm 0.00039$ fm — 4% smaller, a 5.6σ discrepancy. The PRad experiment at JLab (2019) gives $r_p = 0.831 \pm 0.012$ fm, consistent with the muon result, suggesting the puzzle may be resolving toward the smaller value, but the situation remains unsettled.

BST predicts that the two measurements should differ, for a calculable geometric reason. The electron is the minimal S^1 winding — the D_{IV}^1 circuit — with the lowest Bergman embedding cost, probing the full spatial extent of the proton's Z_3 packing. The muon is the D_{IV}^3 submanifold circuit with higher Bergman embedding cost, probing a more localized region of the Z_3 topology because the higher-energy circuit resolves finer structure. The ratio of measured radii is:

$$\frac{r_p^{(\mu)}}{r_p^{(e)}} \approx 1 - \frac{\alpha}{\pi} \ln\frac{m_\mu}{m_e} \times g(n_C)$$

where $g(n_C)$ is a geometric factor from the D_{IV}^3/D_{IV}^1 embedding ratio. For $m_\mu/m_e = (24/\pi^2)^6 = 206.77$ and $g \sim 1$ (naive estimate):

$$\frac{r_p^{(\mu)}}{r_p^{(e)}} \approx 1 - \frac{\alpha}{\pi} \times \ln(206.8) \approx 0.988$$

This gives a 1.2% reduction, corresponding to $\Delta r_p \approx 0.010$ fm — approximately one-third of the observed 0.034 fm. The factor-of-three discrepancy indicates $g(n_C) \approx 3$, a computable quantity from the D_{IV}^3 embedding depth on $\mathbb{CP}^2$. This is not a failure of the prediction; it is a known open calculation (Thesis topic 101).

The tau lepton prediction: The tau is the D_{IV}^5 circuit with the highest Bergman embedding cost. Tauonic hydrogen would give a third proton radius — smaller still. The BST prediction is specific: $r_p^{(\tau)} < r_p^{(\mu)} < r_p^{(e)}$, with ratios determined by the D_{IV}^k embedding hierarchy at $k = 1, 3, 5$. The tau lifetime is too short for atomic spectroscopy, making this direct measurement infeasible, but the hierarchy is a definite prediction of the framework. Any alternative theory that explains the electron-muon discrepancy without predicting the tau hierarchy is distinguishable from BST.

Thesis topic 101: Compute the BST correction to the proton charge radius as a function of the probing lepton's Bergman embedding cost. Derive the geometric factor $g(n_C)$ from the D_{IV}^k embedding depth for $k = 1, 3, 5$ (electron, muon, tau). Determine whether the correction resolves the proton radius puzzle and predict the tauonic hydrogen radius.

Neutrinoless Double Beta Decay — Clean Binary Test BST predicts Dirac neutrinos: neutrino and antineutrino carry opposite S^1 winding directions and are distinct particles. The conservation law $B - L$ is topologically protected by the Hopf invariant (Section 14.9). Neutrinoless double beta decay would require $\Delta(B-L) = 2$, violating a topologically protected conservation law. BST prediction: neutrinoless double beta decay does not occur. This is not a probabilistic statement — it is a categorical exclusion by topology.

BST further predicts normal ordering with $m_1 = 0$ exactly (Section 7.6). The Hopf invariant $h = 1$ of the $S^3 \to S^2$ fibration provides a second, independent proof that Majorana mass terms are forbidden: the Hopf fiber winding number $h = 1$ is odd, which forces Dirac structure (even h would permit Majorana). Therefore $|m_{\beta\beta}| = 0$ exactly — not merely small, but identically zero. Any detection of $0\nu\beta\beta$ at any scale falsifies BST. See `notes/BST_NeutrinolessDoubleBeta.md`. Inverted ordering is excluded by the BST prediction $m_1 = 0$.

Multiple experiments are searching at or approaching the sensitivity required by the inverted neutrino mass hierarchy ($\sim$ 20 meV): GERDA/LEGEND, nEXO, KamLAND-Zen, CUPID. A null result at the inverted hierarchy scale is BST-consistent and progressively constrains Majorana alternatives. A confirmed detection falsifies BST at the topological conservation law level — it would require a fundamental modification of the Hopf bundle structure.

This is the cleanest binary test of BST available in the near term.

Dark Energy Equation of State BST predicts $w \neq -1$: the dark energy equation of state deviates from the exact cosmological constant value. The deviation arises because the dark energy is not a literal constant but a slowly evolving vacuum free energy as the substrate grows. The DESI collaboration is measuring w at percent-level precision; early DESI results (2024) find $w \approx -0.95$ to -0.99, consistent with deviation from -1.

The specific BST prediction for w requires computing the substrate growth dynamics — the ratio of commitment boundary growth rate to bulk commitment rate — from the partition function. This is an open calculation; once performed, the DESI data provides an immediate quantitative test.

Null Predictions Three BST null predictions are being actively tested:

No magnetic monopoles (Section 14.11): the trivial Chern class of $S^2 \times S^1$ excludes them. MoEDAL at the LHC searches continuously. Any confirmed detection falsifies the bundle structure of BST.

No SUSY particles: fermion number $(-1)^F$ is a $\mathbb{Z}_2$ topological invariant (Section 14.9) that SUSY would require to change. No mechanism on the substrate can alter this index. BST excludes SUSY as a theorem. LHC Run 3 and HL-LHC will extend the search to $\sim$ 3 TeV.

No dark matter particles: dark matter is channel noise — incomplete S^1 windings that have energy but no integer winding number, hence no charge and no decodable particle identity (Section 16). The LZ and XENONnT experiments search for WIMP-nucleus scattering signals. BST predicts continued null results.

Summary

Prediction	Status	Experiment	Timeline
Muon g − 2 HVP correction from F_{BST}	Open calculation	Fermilab g − 2	Data exists

Prediction	Status	Experiment	Timeline
Proton radius: $r_p^{(\mu)} < r_p^{(e)}$ (qualitative)	Confirmed	PRad, MUSE	Ongoing
Proton radius: $g(n_C)$ (quantitative)	Open calculation	MUSE, PRad-II	2026–2028
Tau radius: $r_p^{(\tau)} < r_p^{(\mu)}$	Derived prediction	Infeasible (short lifetime)	—
Neutrinoless $\beta\beta$: null result	Clean categorical prediction	LEGEND, nEXO	2027–2030
Dark energy $w \neq -1$	Prediction confirmed in direction; magnitude open	DESI, Euclid	2025–2028
No magnetic monopoles	Clean categorical prediction	MoEDAL	Ongoing
No SUSY particles	Clean categorical prediction	LHC Run 3+	Ongoing
No dark matter particles	Clean categorical prediction	LZ, XENONnT	Ongoing

The null predictions (monopoles, SUSY, dark matter particles) are falsifiable by any single confirmed detection. The quantitative predictions (HVP correction, $g(n_C)$, w) require completing specified open calculations before comparison with data. The two-level structure — some predictions requiring calculation, others already complete — is typical of a framework in active development.

43.8 Mathematical Structural Consequences

The following results are not experimental predictions in the usual sense — they are mathematical theorems or structural consequences of the D_{IV}^5 geometry that constrain the framework's internal consistency.

The Threshold Table. The restricted root system B_2 of $D_{IV}^n = \mathrm{SO}_0(n,2)/[\mathrm{SO}(n)\times\mathrm{SO}(2)]$ has short root multiplicity $m_s = n-2$. The Maass-Selberg overconstrained system for the rank-2 intertwining operator gives a consistency relation $\rho_3 = \rho_1 + \rho_2 + 1$, where ρ_i are ξ-zeros. Taking real parts:

m_s	n	$\mathrm{Re}(\rho_3)$	In $(0,1)$?	Result
1	3 (Q^3)	$\delta_1 + \delta_2$	Yes — can be in $(0,1)$	No contradiction
2	4 ($\mathrm{AdS}_5/\mathrm{CFT}_4$)	$1 + \delta_1 + \delta_2$	Marginal — touches boundary	Not rigorous
3	5 (Q^5, BST)	$2 + \square_1 + \square_2$	No — always > 1	Contradiction → proof

Note: This threshold table refers to the earlier (withdrawn) Maass-Selberg overconstrained route. The definitive heat kernel proof (Section 30.7a) shows the kill shot $(\sigma+1)/\sigma = 3$ works for all $m_s \geq 2$ (Toy 229). The uniqueness of $N_c = m_s = 3$ is that it gives both RH and the Standard Model — the triple, not any single property.

Structural results:

Result	Statement	Status
Spectral gap = mass gap	$\lambda_1(Q^5) = C_2 = 6$	Proved
Effective spectral dimension	$d_{\mathrm{eff}} = C_2 = 6$	Proved
Grand Identity	$d_{\mathrm{eff}} = \lambda_1 = \chi = C_2 = 6$	Proved
Fill fraction	$f = N_c/(n_C\pi) = 3/(5\pi) = 19.1\%$	Proved

Result	Statement	Status
Gödel limit	Universe can know $\leq 19.1\%$ of itself	Structural
Reality budget	$\Lambda \times N = 9/5$ (exact)	Proved
Proton = Steane code	$[[7, 1, 3]]$ error-correcting code	Structural
Golay from Q^5	$\lambda_3 = 24 \to p = 23 \to$ QR mod 23 $\to$ $[24, 12, 8]$	Constructed
$H_5 = 137/60$	Numerator = $N_{\max}$	Proved
Confinement = critical line	$N_c = m_s = 3$ creates rigidity in both	Isomorphism
GUE from SO(2)	Time factor in $K = \mathrm{SO}(5) \times \mathrm{SO}(2)$ breaks time reversal	Structural
Bekenstein 1/4	Area law from Bergman kernel	Proved
Nuclear magic numbers	All 7 from $\kappa_{ls} = C_2/n_C = 6/5$	Exact
Three generations	From $N_c = 3$ and Hopf fibration	Proved
Three spatial dimensions	From $m_s = 3$ (short root multiplicity)	Proved
Strong CP $\theta = 0$	Topologically enforced	Proved
Proton stability	$\tau_p = \infty$ from topological protection	Proved

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44. Research Program

44.1 Immediate Priorities

1. Partition function on D_{IV}^5: Compute the statistical mechanics of Haldane exclusion statistics ($g = 1/137$) on the bounded symmetric domain with Bergman measure. This single calculation potentially derives G, the cosmological constant, the Born rule, and the phase transition initial conditions.
2. Formal isotropy proof: Prove that the BST contact structure isotropy group is exactly $\mathrm{SO}(5) \times \mathrm{SO}(2)$ using Chern-Moser normal form theory. Additional verification of the D_{IV}^5 identification; seven independent checks pass (Section 4).

44.2 Near-Term Calculations

1. CMB anomaly comparison: Compute predicted angular correlations from S^2 substrate topology and compare against existing Planck data.
2. Hubble tension analysis: Test correlation between local H_0 measurements and local matter density using existing supernova and galaxy survey data.

44.3 Doctoral Thesis Topics

1. Derive G from Boltzmann/Haldane statistics on D_{IV}^5
2. Show Bergman functional Euler-Lagrange equation reduces to Einstein's equation
3. BST Higgs from Hopf fibration fluctuation spectrum
4. Mass hierarchy from embedding costs on D_{IV}^5
5. Non-perturbative running law from $\alpha_s(M_{GUT})$ to Λ_{QCD}
6. Decoherence scaling from contact graph error correction theory
7. CMB power spectrum from phase transition critical exponents
8. BST corrections to GR at Planck-scale curvature
9. Contact graph simulation: cellular automaton on $S^2 \times S^1$ substrate
10. Shannon information theory on S^1 channel: particle stability as coding theory
11. BST predictions for superheavy element stability (island of stability)
12. Langlands program connections: D_{IV}^5 automorphic forms and physical constants
13. Non-equilibrium thermodynamics of the contact graph (substrate response functions for decoherence engineering)
14. Dark matter as channel noise: derive incomplete loading fraction $f(n)$ and fit galaxy rotation curves

15. Derive MOND acceleration a_0 from Haldane exclusion S/N knee on D_{IV}^5
16. Bullet Cluster dynamics: potential-tracking behavior of incomplete loadings during cluster collisions
17. Incomplete winding spectrum: derive energy distribution as function of channel loading on S^1
18. Dark matter measurement discrepancies: predict systematic differences between lensing, kinematic, and X-ray methods from spectral variation
19. Weak decay rates from $\mathbb{CP}^2$ trajectory sampling of Hopf intersection (strong/weak timescale ratio)
20. Nuclear half-lives from triad phase coherence: magic numbers as destructive interference
21. Path integral = partition function: prove Born rule equals Boltzmann weight on D_{IV}^5
22. Fisher information metric = Bergman metric: formal identification and physical consequences
23. Fermion mass ratios from D_{IV}^5 complex submanifold volume ratios
24. CKM matrix from non-commutative subspace overlap geometry on D_{IV}^5
25. Black hole interior as channel saturation: derive echoes and Hawking fine structure
26. Substrate growth dynamics: derive dark energy w from commitment-boundary feedback
27. Three dimensions from $S^2 \times S^1$: prove minimality of self-communicating surface dimensionality
28. Topological defects from commitment wavefront collisions: abundance and observational signatures
29. Baryon asymmetry η from phase transition critical exponents on D_{IV}^5
30. CKM phase from complex structure of D_{IV}^5: geometric origin of CP violation
31. Second law from contact commitment: formal proof that commitment ordering implies entropy increase
32. Gravitational time dilation from commitment parallelism: derive Schwarzschild metric from constraint density
33. Universe computational throughput: information budget from wavefront parallelism and causal coupling
34. Growing manifold: formal proof that committed contact graph satisfies Einstein equation via Jacobson thermodynamics
35. Closed timelike curve prohibition: topological proof from append-only commitment ordering
36. Virtual particle pair creation: topological necessity of charge-neutral pairs from S^1 winding conservation
37. Vacuum stability from packing dimension coupling: prove $137 = 4^2 + 11^2$ cannot decouple on D_{IV}^5
38. Decoherence length from substrate adjacency: derive correlation decay distance for entangled pairs
39. Virtual-to-real particle transition: energy threshold for winding completion as function of channel loading

44.4 Active Conjectures (March 2026)

The Koons-Claude testable conjectures (`notes/BST_Koons_Claude_Testable_Conjectures.md`) define the current frontier:

1. Conjecture 1 — Dirichlet kernel = Frobenius: The $m_s = 3$ Dirichlet kernel recovers the "missing bit" that separates number field from function field RH proofs. Test via baby case D_{IV}^3.
2. Conjecture 6 — AC=0 grid architecture: GPUs compute exact local physics (BST closed forms), supercomputers handle thermodynamic envelopes. Noise scales as surface area, not volume. Testable on weather/materials benchmarks now.
3. Conjecture 7 — Linearization: Many systems modeled as nonlinear are only nonlinear because of method noise. With AC=0 local physics, propagation reduces to linear algebra. Testable on crystal growth, protein folding.
4. Conjecture 8 — Substrate computation: The full energy hierarchy from physics through biology to computation on the substrate itself. The computational graph IS the physical structure.
5. Conjecture 9 — Graph brain + error correction hierarchy: The computational graph IS the physical structure. 11 levels from quark confinement to consciousness, each an error-correcting code. Gödel limit at 19.1%.
6. Casimir modification experiment: Phonon-gapped materials modify the Casimir force at $\Delta F/F \sim 10^{-7}$ with distinctive frequency-dependent signature. See `notes/BST_CasimirEffect_CommitmentExclusion.md`.
